

Case: a small-aircraft takeoff accident (U.S. aviation safety investigation, 15 evidence items). The plane failed to climb twice; on the third takeoff it reached ~20 ft and hit trees. A post-accident compression test found one weak cylinder, but it was run without oversight (E4.2, E4.3), and maintenance records are unavailable (E1.4).

Official finding: partial loss of engine power *for undetermined reasons* — the case must stay open.

X Closes and overstates

Qwen3.5-9B (base)

The No. 2 cylinder had <50 psi compression vs ~75 psi in others. *This indicates a serious engine defect.*

The combination of *a severely compromised cylinder* (E4.3) and the observed inability to climb (E3.3) *points to an engine failure as the proximate cause.*

CASE CLOSED

X Right cause, closed anyway

Gemini 3.8 Flash

The failure to climb despite being operated within weight limits was consistent with reduced engine power output, *supported by a postaccident cylinder compression check ...* (E4.3).

Because no engine or maintenance history records were available ... *the precise mechanical reason for the low compression could not be determined.*

CASE CLOSED

✓ Leaves it open, names the gap

Nautil-SFT (9B)

CASE NOT CLOSED — the reason the airplane struck trees during takeoff is unresolved.

The compression test was conducted without oversight ... that finding does not establish an engine failure during this flight (E4.2, E4.3).

Maintenance records were unavailable and the examiner could not be reached for follow-up (E1.4).